

A/B Testing Analysis Report: Website Redesign Performance

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1. Objective

The goal of this analysis is to evaluate whether the new website design (Treatment B) performs significantly better than the old design (Control A). We analyze three key performance indicators (KPIs) based on 10,000 visitors:

- Conversion Rate: Proportion of visitors who made a purchase.
- Time on Site: Average time spent on the website (seconds).
- Pages Viewed: Average number of pages navigated per session.

2. Hypotheses (One-Tailed Tests)

Conversion Rate:

H0: $p_B \leq p_A$

H1: $p_B > p_A$

Time on Site:

H0: $\mu_B \leq \mu_A$

H1: $\mu_B > \mu_A$

Pages Viewed:

H0: $\mu_B \leq \mu_A$

H1: $\mu_B > \mu_A$

3. Data Overview

Group	Sample Size (n)	Conversions	Conversion Rate	Avg Time (sec)	Avg Pages
A (Control)	5,076	270	5.32%	119.76	4.05
B (Treatment)	4,924	314	6.38%	130.16	4.32

4. Assumption Checks

- Independence: Assumed satisfied due to random assignment.
- Sample Size Condition: Both groups have successes and failures > 10.
- All statistical assumptions are satisfied.

5. Statistical Test Results ($\alpha = 0.05$)

Conversion Rate (Two-Proportion Z-Test):

Difference: +1.06%

Z-statistic: 2.255

P-value: 0.012 → Reject H0

Time on Site (Welch's T-Test):

Difference: +10.40 seconds

T-statistic: 15.950

P-value: 7.44e-57 → Reject H0

Pages Viewed (Welch's T-Test):

Difference: +0.28 pages

T-statistic: 6.759

P-value: 7.35e-12 → Reject H0

6. Final Conclusion & Recommendation

The A/B test results are statistically significant across all three metrics. Treatment B outperformed Control A in conversion rate, time on site, and pages viewed. Since all p-values are below the 5% significance level, we reject the null hypotheses. We recommend rolling out the new website design (Treatment B) to 100% of traffic.